

Phase 1: Project Proposal (Topic & Physics Modeling)

1. Student Information

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Selected Topic Type: Inverse

2. Physical System Selection (Mandatory Requirements)

Physical system

The physical system considered in this project is a robot joint motor housing operating under time-varying electrical current. During motor operation, electrical losses in the winding and nearby internal components generate heat through Joule heating. This heat is conducted through the motor housing and dissipated to the surrounding air by convection.

Since excessive motor temperature can reduce efficiency, damage insulation, and affect the reliability of a robotic joint, estimating the thermal behavior of the motor housing is an important engineering problem. Therefore, this project focuses on predicting the transient temperature distribution in the motor housing and estimating unknown thermal parameters using a physics-informed neural network, PINN.

Geometry

To make the problem suitable for a PINN-based heat transfer study, the motor housing is simplified as a one-dimensional cylindrical body along the axial direction. The temperature field is assumed to be cross-sectionally averaged, so the main unknown temperature field is

$$T(x, t)$$

where x is the axial coordinate of the motor housing and t is time. The housing length is denoted by L , and the computational domain is

$$0 \leq x \leq L$$

A representative geometry is selected as follows:

$$L = 0.15 \text{ m}, R_o = 0.04 \text{ m}, R_i = 0.025 \text{ m}$$

where R_o is the outer radius and R_i is the inner radius of the cylindrical motor housing.

The effective cross-sectional area for axial conduction is

$$A = \pi(R_o^2 - R_i^2),$$

and the outer perimeter exposed to air is

$$P = 2\pi R_o.$$

This simplified geometry allows the motor housing to be modeled as a one-dimensional transient conduction-convection system while still retaining the major physical effects of heat generation, axial conduction, and convective cooling.

Material Properties

The material of the motor housing is assumed to be aluminum alloy. Aluminum alloy is commonly used for motor casings because of its relatively high thermal conductivity, low weight, and good manufacturability.

The material properties used in this project are

$$\rho = 2700 \text{ kg/m}^3, c_p = 900 \text{ J/(kg}\cdot\text{K)}, k = 167 \text{ W/(m}\cdot\text{K)}.$$

Here, ρ is the density, c_p is the specific heat, and k is the thermal conductivity of the motor housing material.

The ambient temperature is assumed to be $T_\infty = 25^\circ\text{C}$.

The convective heat transfer coefficient h is treated as an unknown parameter to be estimated because it depends on operating conditions such as airflow, motor orientation, and the surrounding environment. The Joule heating coefficient q_0 is also treated as an unknown or partially unknown parameter because the exact internal electrical loss distribution is difficult to measure directly.

The inner surface of the housing faces the motor internals (rotor, windings) and is not directly exposed to the ambient air. Therefore, only the outer cylindrical surface participates in convective heat loss, and the effective convection perimeter is taken as $P = 2\pi R_o$.

Chosen Complexity

The project includes four important complexities: Transient heat transfer, Internal Joule heat generation, Combined conduction and convective cooling, Inverse parameter estimation

This complexity was chosen because it is more realistic than a simple steady-state conduction problem and better represents the thermal behavior of an actual robot joint

motor. During operation, the motor temperature changes with time as the electrical current input changes. Therefore, transient heat transfer must be considered. In addition, heat is generated internally by electrical losses and dissipated to the surrounding air by convection, so both heating and cooling effects must be included.

The inverse problem formulation is also important because some thermal parameters are difficult to know accurately in practice. For example, the convective heat transfer coefficient h depends on airflow, motor orientation, and the surrounding environment. Similarly, the Joule heating coefficient q_0 depends on the internal electrical loss distribution. Instead of assuming these parameters are exactly known, this project aims to estimate them from sparse temperature sensor data using a PINN.

Therefore, this problem satisfies the requirements of a non-trivial heat transfer project involving transient behavior, internal heat generation, conduction-convection coupling, and inverse parameter estimation.

3. Governing Equations & Boundary Conditions

Governing Equation

The robot joint motor housing is modeled as a one-dimensional transient conduction-convection system with internal Joule heat generation. The temperature is assumed to be cross-sectionally averaged, so the temperature field is expressed as

$$T = T(x, t)$$

where x is the axial coordinate of the motor housing and t is time.

The governing heat transfer equation is

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} - \frac{hP}{A} (T - T_\infty) + q_0 I(t)^2 g(x)$$

In this equation, ρ is the density, c_p is the specific heat, k is the thermal conductivity, h is the convective heat transfer coefficient, P is the outer perimeter exposed to air, A is the effective cross-sectional area, T_∞ is the ambient temperature, $I(t)$ is the time-dependent motor current, $g(x)$ is the spatial heat generation distribution, and q_0 is the Joule heating coefficient.

In this project, $I(t)$ is treated as a known input signal measured from the motor controller. The input current is assumed to include both current-on and current-off intervals. During current-on intervals, $q_0 I(t)^2 g(x)$ drives Joule heating, while during current-off intervals, $I(t) = 0$, so the Joule heating term vanishes and the cooling response becomes mainly sensitive to h .

The physical meaning of each term

- (1) The transient storage term: $\rho c_p \frac{\partial T}{\partial t}$

This term represents the rate of thermal energy storage inside the motor housing. It is necessary because the motor current changes with time, causing the temperature field to evolve transiently.

- (2) The axial conduction term: $k \frac{\partial^2 T}{\partial x^2}$

This term represents heat conduction along the axial direction of the motor housing.

- (3) The convective cooling term: $-\frac{hP}{A}(T - T_\infty)$

This term represents heat loss from the outer cylindrical surface of the motor housing to the surrounding air. Since the original cylindrical surface is reduced to a one-dimensional cross-sectionally averaged model, the lateral convection effect appears as a distributed cooling sink term in the governing equation instead of a separate boundary condition.

- (4) The internal Joule heating term: $q_0 I(t)^2 g(x)$

This term represents heat generation caused by electrical losses in the motor winding and nearby internal components. The current dependence is modeled as proportional to $I(t)^2$, which is consistent with Joule heating. The function $g(x)$ describes the spatial distribution of Joule heat generation. In this project, $g(x)$ is modeled as a Gaussian distribution concentrated near the motor winding region, as defined in the section below.

Initial Condition

The motor housing is assumed to start from a uniform ambient temperature:

$$T(x, 0) = T_\infty$$

This initial condition represents the motor before operation, when the entire housing is at thermal equilibrium with the surrounding air

In this project, the spatial heat generation distribution is modeled as a Gaussian function centered at the motor winding location:

$$g(x) = \exp\left(-\frac{(x - x_0)^2}{2\sigma^2}\right)$$

where $x_0 = 0.3L = 0.045$ m is the center of the motor winding region, and $\sigma = 0.1L = 0.015$ m controls the spatial extent of the heat source. This concentrates Joule heating near the winding zone and produces a spatially non-uniform temperature field, which is essential for meaningful sensor placement in the inverse problem.

Boundary Conditions

The axial ends of the simplified one-dimensional motor housing are assumed to be adiabatic. Therefore, no heat flux leaves through the two axial ends.

$$\text{At } x = 0: \frac{\partial T}{\partial x}(0, t) = 0$$

$$\text{At } x = L: \frac{\partial T}{\partial x}(L, t) = 0$$

These boundary conditions mean that the dominant heat loss mechanism is assumed to be convection from the outer cylindrical surface, which is already included in the governing equation through the distributed cooling term

$$-\frac{hP}{A}(T - T_\infty)$$

This modeling choice keeps the problem one-dimensional while still including the combined effects of axial conduction, lateral convective cooling, and internal Joule heat generation.

PDE Residual for PINN

For the PINN formulation, the governing equation is rearranged into a residual form. The PDE residual is defined as

$$f_{PDE}(x, t) = \rho c_p \frac{\partial T_\phi}{\partial t} - k \frac{\partial^2 T_\phi}{\partial x^2} + \frac{hP}{A}(T_\phi - T_\infty) - q_0 I(t)^2 g(x)$$

where $T_\phi(x, t)$ is the neural network prediction of the temperature field.

The PDE loss is then defined as the mean squared residual over collocation points:

$$\mathcal{L}_{PDE} = \frac{1}{N_f} \sum_{i=1}^{N_f} |f_{PDE}(x_i, t_i)|^2$$

The boundary condition loss is defined as

$$\mathcal{L}_{BC} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| \frac{\partial T_\phi}{\partial x}(0, t_i) \right|^2 + \frac{1}{N_b} \sum_{i=1}^{N_b} \left| \frac{\partial T_\phi}{\partial x}(L, t_i) \right|^2$$

The initial condition loss is

$$\mathcal{L}_{IC} = \frac{1}{N_0} \sum_{i=1}^{N_0} |T_\phi(x_i, 0) - T_\infty|^2$$

For the inverse problem, sparse temperature sensor data are also included through a data loss:

$$\mathcal{L}_{data} = \frac{1}{N_s} \sum_{i=1}^{N_s} |T_\phi(x_i, t_i) - T_{\text{sensor},i}|^2$$

Therefore, the total PINN loss is

$$\mathcal{L}_{total} = w_{PDE} \mathcal{L}_{PDE} + w_{IC} \mathcal{L}_{IC} + w_{BC} \mathcal{L}_{BC} + w_{data} \mathcal{L}_{data}$$

In the inverse PINN setting, unknown parameters such as the convective heat transfer coefficient h and the Joule heating coefficient q_0 can be treated as trainable variables. The PINN then learns both the temperature field $T(x, t)$ and the unknown thermal parameters by minimizing the total loss. In practice, all loss terms are evaluated in the nondimensional domain using θ, X, τ as defined in the Scaling and Normalization section below, and the unknown parameters are learned as the dimensionless quantities β and S , then converted back to the physical parameters h and q_0 after training.

Scaling and Normalization

Directly training the PINN using SI units may cause scaling issues because the magnitudes of x, t , temperature, and material properties can be very different. For example, $k, \rho c_p, hP/A$, and $q_0 l(t)^2$ may have significantly different numerical scales. This can make the PDE loss difficult to balance with the boundary and data losses. To improve training stability, nondimensional variables are introduced:

$$X = \frac{x}{L}, \tau = \frac{\alpha t}{L^2}, \theta = \frac{T - T_\infty}{\Delta T_{\text{ref}}}$$

where $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity, and ΔT_{ref} is a reference temperature difference.

The motor current is also normalized as $i(\tau) = \frac{I(t)}{I_{\text{ref}}}$.

Using these variables, the governing equation can be written in nondimensional form as

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial X^2} - \beta \theta + S i(\tau)^2 g(X)$$

where $\beta = \frac{hPL^2}{kA}$ is the nondimensional convective cooling parameter, and $S = \frac{q_0 I_{\text{ref}}^2 L^2}{k \Delta T_{\text{ref}}}$ is the nondimensional Joule heating parameter.

This nondimensional form is expected to improve convergence because all input and output variables are kept near order one. In the PINN implementation, the inverse parameters may be learned as β and S , and then converted back to the physical parameters h and q_0 after training.

★ [BONUS/ADVANCED] Inverse Problem Design (+Extra Credit)

Sensor Placement Strategy

Because the heat generation distribution is modeled as a Gaussian centered at $x_0 = 0.3L$, the temperature field is spatially non-uniform. This allows sensors at different axial positions to observe physically distinct thermal responses, enabling spatial identifiability of the two unknown parameters. Three sparse temperature sensors are placed at the following nondimensional locations:

$$X_1 = 0.25, X_2 = 0.55, X_3 = 0.85$$

Physical sensitivity of each sensor location:

(1) Sensor S1 at $X_1 = 0.25$ — sensitive to S (Joule heating coefficient)

This sensor is placed just inside the Gaussian peak ($X_0 = 0.3$), where heat generation is near its maximum value. During current-on periods, the temperature rise at S1 is the steepest of all sensors, making it the most informative location for identifying the Joule heating parameter S (and hence q_0).

(2) Sensor S2 at $X_2 = 0.55$ — full-field reconstruction

This sensor is located in the intermediate conduction zone, where neither heat generation nor convective cooling dominates alone. It observes how heat diffuses axially from the winding region toward the cooler far end, helping the PINN reconstruct the full spatial temperature profile. It also provides an independent constraint that prevents degenerate parameter combinations that could otherwise fit S1 and S3 data alone.

(3) Sensor S3 at $X_3 = 0.85$ — sensitive to β (convection coefficient)

This sensor is far from the Gaussian heat source, where the local heat generation value $g(0.85) \approx 0$ is negligible. The local temperature is governed primarily by the balance between incoming axial conduction and outgoing lateral convection. A larger h (larger β) reduces the temperature at S3 more strongly than at S1, making this sensor the most informative location for estimating β and hence h .

Temporal identifiability from the current profile:

Spatial sensor placement alone is not sufficient. The two unknowns must also be separable in the time domain. To achieve this, the motor current is designed with distinct heating and cooling phases. During current-on intervals, the Joule heating term $q_0 I^{(t)2} g(x)$ is active and sensor S1 near the Gaussian peak provides the strongest signal for estimating q_0 . During current-off intervals, $I(t) = 0$, the Joule term vanishes and the temperature decays purely under convective cooling. The decay rate is governed exclusively by h , and sensor S3 at the cooling-dominated far end provides the cleanest estimate of h .

The combination of spatially distinct sensor locations (exploiting the Gaussian shape of $g(x)$) and temporally distinct current phases (separating the influence of the two unknowns) ensures that both inverse parameters are identifiable from the sparse sensor data.